Week 5 (3/23-3/29, 2026)

We mainly did some coding/plotting this week.

stack AHF output orbit + galpy predicted orbit

open & read a file in python:

> W3Schools.com
> W3Schools offers free online tutorials, references and exercises in all the major languages of the web. Covering popular subjects like HTML, CSS, JavaScript, Python, SQL,
> https://www.w3schools.com/python/python_file_open.asp



`.strip()` function in python:

> W3Schools.com
> W3Schools offers free online tutorials, references and exercises in all the major languages of the web. Covering popular subjects like HTML, CSS, JavaScript, Python, SQL,
> https://www.w3schools.com/python/ref_string_strip.asp



`.split()` function in python:

> W3Schools.com
> W3Schools offers free online tutorials, references and exercises in all the major languages of the web. Covering popular subjects like HTML, CSS, JavaScript, Python, SQL,
> https://www.w3schools.com/python/ref_string_split.asp



`.readlines()` function in python:

W3Schools.com

W3Schools offers free online tutorials, references and exercises in all the major languages of the web. Covering popular subjects like HTML, CSS, JavaScript, Python, SQL,

https://www.w3schools.com/python/ref_file_readlines.asp



`float()` function in python:

W3Schools.com

W3Schools offers free online tutorials, references and exercises in all the major languages of the web. Covering popular subjects like HTML, CSS, JavaScript, Python, SQL,

https://www.w3schools.com/python/ref_func_float.asp



`.append()` function for list in python:

W3Schools.com

W3Schools offers free online tutorials, references and exercises in all the major languages of the web. Covering popular subjects like HTML, CSS, JavaScript, Python, SQL,

https://www.w3schools.com/python/ref_list_append.asp



3D plot with matplotlib:

mpl_toolkits.mplot3d.axes3d.Axes3D.plot — Matplotlib 3.10.8 documentation

z coordinates of vertices; either one for all points or one for each point.

https://matplotlib.org/stable/api/_as_gen/mpl_toolkits.mplot3d.axes3d.Axes3D.plot.html

make density profile plot for one single snapshot

logic of code:

create bins array on logscale → create bin centers array → build histogram for total mass in each bin → divide each total mass by the volume for each bin → plot

density vs bin center

- `numpy.logspace()`: Return numbers spaced evenly on a log scale.

```
numpy.logspace(start, stop, num=50, endpoint=True, base=10.0, dtype=None, axis=0)[source]
```

> numpy.logspace — NumPy v2.5.dev0 Manual
> In linear space, the sequence starts at base ** start
> (base to the power of start) and ends with base ** stop
> (see endpoint below).
> https://numpy.org/devdocs/reference/generated/numpy.logspace.html

- `numpy.histogram()`: Compute the histogram of a dataset.

```
numpy.histogram(a, bins=10, range=None, density=None, weights=None)[source]
```

> numpy.histogram — NumPy v2.5.dev0 Manual
> If bins is an int, it defines the number of equal-width
> bins in the given range (10, by default). If bins is a
> sequence, it defines a monotonically increasing array of bin edges,
> https://numpy.org/devdocs/reference/generated/numpy.histogram.html

- `numpy.linalg.norm()`: Matrix or vector norm.

> numpy.linalg.norm — NumPy v2.5.dev0 Manual
> This function is able to return one of eight different matrix norms,
> or one of an infinite number of vector norms (described below), depending
> on the value of the ord parameter.
> https://numpy.org/devdocs/reference/generated/numpy.linalg.norm.html

- `numpy.log10()`: Return the base 10 logarithm of the input array, element-wise.

numpy.log10 — NumPy v2.1 Manual

A location into which the result is stored. If provided, it must have a shape that the inputs broadcast to. If not provided or None, a freshly-allocated array is returned. A tuple (possible only as a

https://numpy.org/doc/2.1/reference/generated/numpy.log10.html

- `numpy.sqrt()`: Return the non-negative square-root of an array, element-wise.

numpy.sqrt — NumPy v2.2 Manual

A location into which the result is stored. If provided, it must have a shape that the inputs broadcast to. If not provided or None, a freshly-allocated array is returned. A tuple (possible only as a

https://numpy.org/doc/2.2/reference/generated/numpy.sqrt.html

- `os.path.join()`: Join one or more path segments intelligently.

os.path — Common pathname manipulations

Source code: Lib/genericpath.py, Lib/posixpath.py(for POSIX) and Lib/ntpath.py(for Windows). This module implements some useful functions on pathnames. To read or write files see

https://docs.python.org/3/library/os.path.html#os.path.join

